Mechanics of Soft Materials (2025) 7:5
https://doi.org/10.1007/s42558-025-00065-8

REVIEW PAPER

Cavitation rheology: advances in soft material mechanics and biomechanical applications

Daniel Coen[1] · **Kaylin Shanahan**[1] · **Wanis Nafo**[2]

Received: 31 March 2025 / Accepted: 24 August 2025 / Published online: 28 November 2025
© The Author(s), under exclusive licence to Springer Nature Switzerland AG 2025

Abstract
Cavitation rheology has rapidly established itself as a crucial methodology for probing the mechanical properties of soft materials and biological tissues. This review examines the method's principles, techniques, and applications within the soft material mechanics domain. By addressing the inherent limitations of traditional bulk testing methods, cavitation rheology provides unparalleled precision in characterizing material responses. The discussion includes advancements in mathematical modeling and computational techniques that refine the interpretation of cavitation data, highlighting the synergy between experimental accuracy and theoretical modeling. We explore the integration of cavitation rheology with other biomechanical approaches, such as tensile testing and shear wave elastography, to demonstrate the methodology's potential to link localized measurements with bulk mechanical properties. Tissue-specific applications, methodological challenges, and advancements in tissue engineering, oncology, and neurology are also assessed. This synthesis of recent progress emphasizes the transformative role of cavitation rheology in biomechanics, offering significant promise for both experimental and translational applications in diagnostics and therapeutics.

Keywords Cavitation rheology · Soft material mechanics · Tissue engineering · Computational modeling · Biomechanical characterization

1 Introduction

The mechanical behavior of soft materials and biological tissues under stress constitutes a pivotal domain of study in biomechanics, with broad implications for biomedical science and engineering. The response of these materials to mechanical forces is critical to applications spanning tissue engineering, neurology, oncology, and regenerative medicine [1–3]. Among the methodologies gaining traction for their precision and adaptability, cavitation rheology has emerged as particularly promising. It enables a detailed exploration of the mechanical behavior of soft and heterogeneous materials, linking experimental observations with theoretical models.

Cavitation rheology distinguishes itself by its ability to probe localized mechanical responses, a crucial potential when dealing with soft biological tissues or engineered biomaterials. Traditional bulk mechanical testing methods often struggle to capture the complex details of material deformation at localized scales. By inducing a controlled cavity within the medium and measuring the critical pressure required for its growth or collapse, this methodology provides a precise measure of properties such as stiffness, elasticity, and fracture toughness [4, 5]. This localized assessment is invaluable for materials where heterogeneity and strain-dependent behaviors are prominent, as seen in hydrogels and biological tissues, including brain matter, lungs, liver, and their gel-based phantoms. These materials often exhibit nonlinear stress–strain behavior, viscoelasticity, and anisotropy: characteristics that challenge conventional rheological techniques [6]. Although hydrogels

✉ Wanis Nafo
Wanis.Nafo@xjtlu.edu.cn

[1] School of Physics, University of Galway, Galway University Road, Galway H91 CF50, Ireland

[2] School of Mathematical and Statistical Sciences, University Road, Galway H91 TK33, Ireland

demonstrate relatively uniform responses to cavity expansion, making them a model system for studying time-dependent viscoelasticity [7], biological tissues present a significant challenge due to their structural heterogeneity and susceptibility to damage during testing [8]. Cavitation rheology addresses these challenges by modifying experimental parameters, such as cavity size and deformation rate, to match the specific properties of the material being tested [9].

Despite its versatility, cavitation rheology is not without limitations. Key challenges include the influence of boundary conditions and assumptions about material isotropy. Studies have shown that boundary effects can significantly alter stress distributions near cavity walls, leading to potential underestimation of material properties. Furthermore, assumptions of isotropy in soft materials, which often exhibit anisotropic or orthotropic mechanical properties, can limit the accuracy of current models. Advances in imaging techniques and computational modeling are essential for addressing these challenges and enabling more precise characterization of complex behaviors, which this review will explore.

The integration of computational simulations with experimental cavitation rheology has been transformative. Computational tools such as finite element methods (FEM) provide a framework for modeling complex material behaviors that are difficult to capture experimentally. For instance, Mishra et al. [10], investigated the influence of needle size and elasto-capillary numbers on cavitation thresholds, facilitating optimized experimental design. Such simulations allow researchers to explore a wide range of conditions, reducing the time and cost associated with physical trials while enhancing the interpretability of experimental data. Moreover, computational modeling enables the development of advanced constitutive equations that capture nonlinear and time-dependent material behaviors. Studies implementing hyperelastic (e.g., Ogden, Yeoh, and Arruda–Boyce) frameworks have demonstrated their efficacy in describing strain-stiffening responses, particularly for soft tissues and polymers [11]. Such advancements not only improve the accuracy of cavitation rheology measurements but also pave the way for its application in increasingly complex materials and conditions. For instance, the implications of cavitation rheology extend far beyond the laboratory. Specifically, in tissue engineering, the ability to characterize the mechanical properties of scaffolds is critical for ensuring their compatibility with native tissues. This is evident in Zimberlin et al. [12], who demonstrated the use of cavitation rheology to evaluate the mechanical behavior of biomaterials designed for scaffold applications, while other researchers have explored its potential in direct medical fields such as neurology and oncology [13, 14].

In this article, the term "cavitation rheology" is employed as an umbrella term for investigating the mechanics of soft materials through cavity expansion phenomena. As detailed in the sections that follow, quasi-static cavity expansion is typically achieved via fluid injection through needles, while dynamic loading conditions are primarily driven by laser-induced cavitation. Although other forms of cavitation, such as acoustic and shockwave-induced, have been identified, their rheological applications remain largely underexplored. For a comprehensive review of these techniques, see Barney et al. [15]. This review focuses on the rheological applications of quasi-static and laser-induced cavitation.

This review examines cavitation rheology, focusing on its principles, methodologies, and applications in soft material mechanics. The discussion is structured to progress logically from foundational concepts to advanced applications, encompassing several key themes. These include the mathematical frameworks that form the basis of cavitation rheology, the integration of computational simulations to refine experimental accuracy, and the challenges and limitations inherent in current experimental setups. Moreover, the review explores tissue-specific responses to cavitation, highlighting the factors that influence these behaviors, and discusses the synergistic potential of combining cavitation rheology with other biomechanical testing techniques. Clinical and biomedical applications, particularly in tissue engineering, oncology, and neurology, are also discussed, emphasizing the method's translational potential. Finally, future research directions are outlined to address current limitations and broaden the scope of cavitation rheology applications. By synthesizing recent advancements and identifying critical gaps in the field, this review aims to serve as a valuable resource for researchers and clinicians alike, facilitating both experimental and practical implementations.

2 Fundamentals and biomechanical applications of cavitation rheology

The core principles of cavitation rheology involve understanding the initiation, growth, and collapse of cavities induced by applied pressure. This process depends on factors such as the material's viscosity and elasticity. By quantifying the critical pressure at which mechanical instability occurs, cavitation rheology determines the material's modulus, enabling the assessment of properties such as stiffness and fracture resistance. Typically, a single cavity is induced in a soft medium, and its critical pressure is measured to evaluate the material's mechanical response [16].

Cavitation rheology is typically implemented through a syringe-based system that injects fluid into a soft material via a fine needle, while continuously monitoring the internal pressure. As illustrated in Fig. 1, the setup consists of a syringe

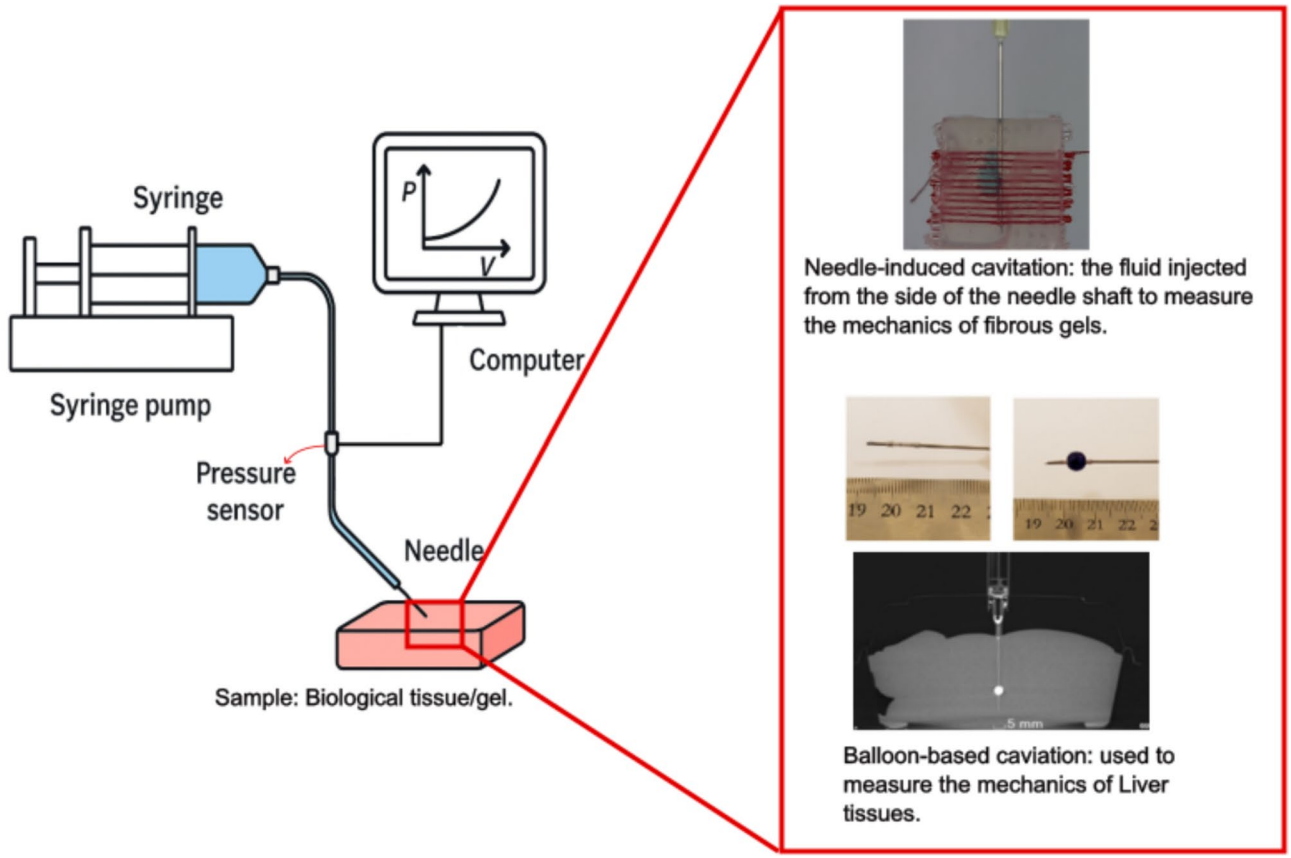


Fig. 1 Schematic of a cavitation rheology experimental setup used to characterize the mechanical properties of soft materials and biological tissues. A syringe pump delivers fluid at a controlled rate through a needle inserted into the sample, while a pressure sensor continuously records the internal pressure. The pressure–volume data are acquired via a connected computer. The magnified views on the right illustrate two commonly used techniques: (top) NIC, where fluid is injected from a needle into gels; (bottom) BRC, where a balloon expands within tissue such as liver to measure bulk mechanical response. These methods enable localized, minimally invasive assessment of stiffness and time-dependent behavior in soft materials

pump connected to a pressure sensor and a computer-based data acquisition system. The needle is inserted into the soft sample, and fluid is injected at a controlled rate. A sharp increase in pressure followed by a sudden drop indicates the onset of cavitation, a localized mechanical instability that reflects the material's resistance to volumetric deformation. Two common experimental configurations are shown: in needle-induced cavitation (NIC), fluid is commonly delivered at the bevel of the needle, or, as demonstrated in Fig. 1, through a side-port needle to generate a small cavity within fibrous gels; in balloon-regulated cavitation (BRC), a thin-walled balloon expands within a soft organ such as liver tissue, enabling the assessment of bulk viscoelastic response. The pressure data at the moment of cavitation are recorded and interpreted using appropriate mechanical models to estimate local elastic or viscoelastic parameters. This approach enables localized, minimally invasive probing of soft material mechanics under physiologically relevant conditions.

In biomechanics, cavitation rheology provides a critical framework for assessing the mechanical behavior of biological tissues under applied mechanical loads. By probing localized cavitation events, investigators can measure how tissues respond to stress, thereby informing broader questions of tissue functionality and integrity. Such knowledge assumes particular importance in tissue engineering, where precise understanding of mechanical properties underpins the design and optimization of both synthetic and regenerated biological materials. A suite of methodologies has evolved to facilitate these investigations. Prominently, NIC [4, 17–19] and BRC [11, 20] have been widely employed, alongside volume-controlled cavity expansion (VCCE) [21] and laser-induced cavitation (LIC) [22, 23]. Both NIC and BRC, executed under quasi-static conditions, have proven useful for characterizing the time-dependent mechanical behavior of soft polymers and biological tissues [7, 24]. Notably, BRC facilitates the formation of cavities along the needle shaft, an approach that inherently mitigates residual strain effects [25]. As these methods refine our ability to probe mechanical responses, they promise to advance both biomechanical understanding and tissue engineering applications.

By contrast, LIC proves very effective in contexts necessitating high strain-rate loading, given its capacity for delivering concentrated energy at rates on the order of 10^3 and 10^8 s^{-1} [15]. While it has been used in medical and biological applications, its utility in material characterization is largely confined to high-energy, short-duration processes.

3 Cavitation rheology in soft matter mechanics and characterization

A cavity is generated by injecting a prescribed volume of incompressible fluid into the target material and measuring the pressure at which abrupt deformation or collapse occurs. This critical pressure, P_c, provides a direct measure of the local material modulus [16]. It is expressed mathematically as [4, 26]:

$$P_c = P_o + P \tag{1}$$

where P_o is the ambient pressure, r represents the deformed cavity radius, γ is the surface tension of the medium, and P_{elas} accounts for the elastic strain energy of the surrounding material. This equation expresses the balance between internal pressure, surface energy, and elastic restoring energy in an expanding cavity.

Several studies have used this relationship to characterize local material properties, with the Neo-Hookean strain energy function commonly applied to describe mechanical responses in soft solids [3–5, 15]. In this framework, the strain energy density W is given by

$$W = \left(\frac{E}{6}\right)\left(2\lambda^2 + \lambda^{-4} - 3\right) \tag{2}$$

To relate the total internal pressure P to the material's elastic properties and surface effects, we begin with the pressure balance across the cavity surface:

$$P = P_{\text{elas}} + \frac{2\gamma}{r} \tag{3}$$

where P_{elas} is the elastic pressure generated by the surrounding material, and γ is the surface tension. Under spherical symmetry and the assumption of a Neo-Hookean solid, the normalized form of the elastic pressure (as derived in [4, 16]) is given by

$$\frac{P_{elas}}{E} = \frac{5}{6} - \frac{2}{3\lambda} - \frac{1}{6\lambda^4} \tag{4}$$

where $\lambda = \frac{r}{r_o}$ is the cavity stretch ratio. Substituting back into the original pressure balance yields

$$\frac{P - \frac{2\gamma}{r}}{E} = \frac{5}{6} - \frac{2}{3\lambda} - \frac{1}{6\lambda^4} \tag{5}$$

E is the Young's modulus, which correlates linearly with the medium's surface tension.

While this approach provides valuable insights into the deformation mechanics of soft materials, it has two key limitations. First, it presumes isotropic and incompressible behavior, assumptions that do not strictly hold for many soft polymers and biological tissues [27]. Second, mechanical properties derived via cavitation can deviate from those obtained through traditional shear rheology, generally yielding higher measured moduli [28, 29]. Bridging these differences typically requires material-specific correction factors [19, 30].

To accurately capture the material response during cavity expansion, the elastic term in Eq. (1) is often replaced by:

$$\int_1^{\lambda} \frac{W(\lambda)}{(\lambda^3 - 1)} d\lambda \tag{6}$$

as outlined by [31] and [32]. This adaptation accommodates strain energy functions (SEFs) that include strain-stiffening behavior, further refined in later developments [11, 33]. Equation (6) expresses the elastic contribution to the cavity pressure derived under quasi-static spherical expansion, formulated in the reference configuration. Here, $W(\lambda)$ denotes the strain energy density per unit undeformed volume, and the integration proceeds from the reference state ($\lambda = 1$) to the current stretched state (λ). This form follows the thick-shell mechanics framework described in [31].

In soft biological tissues, which are typically probed on a highly localized scale [25], Eq. (6) can be implemented in the differential form:

$$\frac{dP}{d\lambda} = \frac{W}{(\lambda^3 - 1)} \tag{7}$$

thereby enabling calibration of material constants for more complex SEFs (Fig. 2).

While different forms may appear in the literature, for instance, in membrane-based models such as those in [32], where the denominator appears as $1 - \lambda^3$ and the integration direction is reversed, these reflect alternate assumptions (e.g., thin-shell approximation) and integration conventions. Both forms are mathematically consistent when their respective assumptions and sign conventions are accounted for.

It is worth mentioning that under spherical symmetry and incompressibility, the deformation field induced by cavitation is equibiaxial in the circumferential directions, i.e., $\lambda_\Theta = \lambda_\phi = \lambda$. This assumption is consistent with experimental and computational observations in soft hydrogels [20].

In needle-based or balloon-based cavitation rheology, needle insertion creates a channel that initially relieves confinement around the expanding cavity. As a result, the injected volume at early stages primarily displaces the gel along this preformed channel without generating measurable pressure. This behavior has been experimentally addressed in balloon-based cavitation studies, where an initial volume offset is introduced to define the reference configuration for stretch calculations [25]. Specifically, pressure begins to rise only once the balloon expansion exceeds the channel boundary and interacts with the surrounding gel. The stretch λ is then computed relative to this point, ensuring that pressure–stretch data reflect the material's true mechanical response, independent of the initial needle-induced displacement phase. While this empirical strategy reduces the influence of early-stage damage effects, further investigation is required to fully characterize the mechanical consequences of needle insertion in both balloon and direct cavitation rheology setups.

Moving beyond purely elastic considerations, time-dependent behavior in quasi-static and dynamic cavitation experiments is often captured through linear viscoelastic models [7, 22]. In this context, the circumferential stress $\sigma^i_{v.el}$ evolves according to

$$\sigma^i_{v.el}(t + \Delta t) = e^{\frac{\triangle t}{T_i}} \cdot \sigma^i_{v.el}(t) + g_i \cdot \sigma_{h.el}(t)\left[1 - e^{\frac{\triangle t}{T_i}}\right] + g_i \frac{\triangle \sigma_{h.el}}{\triangle t}\left[\triangle t - T_i + T_i e^{\frac{\triangle t}{T_i}}\right] \tag{8}$$

where g_i and τ_i are relaxation parameters from a 3-term Prony series, and $\sigma_{h.el}(t + \Delta t)$ represents the circumferential stresses dominant in cavitation experiments [20]. The viscoelastic stress expression in Eq. (6) is formulated in the current configuration and represents the radial and circumferential Cauchy stresses observed in cavitation tests of PVA hydrogels, as developed in [7].

Under higher strain rates, as in LIC, Estrada et al. [34] proposed a Kelvin–Voigt–based viscoelastic framework to capture the response of hydrogels [34]. In this formulation:

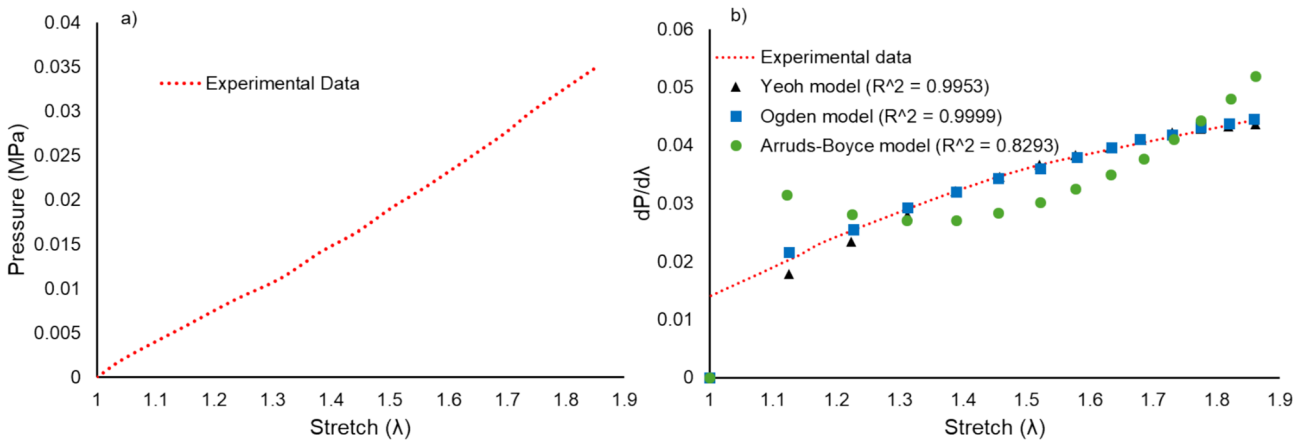


Fig. 2 **a** The experimental data, presented as a stretch term derived from the injected volume, were processed following the methodology detailed in [25]. **b** The instantaneous tangent, calculated from the experimental data throughout the test, serves as the basis for calibrating the hyperelastic models using Eq. (5), adopted with permission from [25]

$$\frac{\mu}{G_1}\dot{S}+S=-\frac{G}{2}[5-\left((\frac{R_0}{R})^4-4\frac{R_0}{R}\right]-\frac{4\mu\dot{R}}{R}\{1+\frac{G}{G_1}\frac{R^3}{R^3-R_0^3}[\frac{3}{14}+\frac{R_0}{R}-\frac{3}{2}(\frac{R_0}{R})^4+\frac{2}{7}\left((\frac{R_0}{R})^7\right]\} \tag{9}$$

where μ and G are material parameters, R is the bubble radius, and S is a stress integral characterized by the radial (pressure) and circumferential stresses S_{rr} *and* S_{ee} generated by the expanding bubble, and expressed as

$$S=\int_R^\infty \frac{2}{r}\left(S_{rr-}S_{\theta\theta}\right)dr \tag{10}$$

Equation (9) describes the dynamic stress response derived under high strain-rate cavitation, also in the current configuration, and its parameters are typically estimated using a least squares (LSQ) fit to initial experimental peaks, assuming a one-parameter model (purely viscous or elastic). If these fail to capture key features such as attenuation or oscillation frequency, a two-parameter Kelvin–Voigt model is applied to refine both modulus G and viscosity μ. In contrast, Eq. (8) evaluates the stress integral in terms of second Piola–Kirchhoff stress components (S_{rr} and S_{ee}) and is thus expressed in the reference configuration. This form is often used in finite element simulations of cavitation dynamics in soft solids.

In complex systems, nonlinear viscoelastic models based on the Standard Linear Solid framework can be introduced to capture additional material behaviors, such as rate-dependent effects associated with parameters such as G_1. In these models, elastic elements are replaced by hyperelastic springs to accommodate large deformations while preserving time-dependent behavior. Convergence criteria and iterative refinements are then used to determine the best-fit model for a given dataset.

While the constitutive models described above are robust for many homogeneous and isotropic materials, the broader methodology remains highly adaptable. Alternative formulations may be readily inserted to account for special cases such as anisotropy, spherical symmetry, or inhomogeneity, thereby expanding the versatility of cavitation rheology for diverse material systems.

4 Common mathematical models and limitations in cavitation rheology

In cavitation rheology, the mechanical properties of soft materials are typically inferred by analyzing pressure–volume data. Although initial studies often relied on a Neo-Hookean framework, a variety of SEFs have subsequently been introduced to enhance predictive accuracy. The following section surveys these models, with particular emphasis on their respective applications and recognized constraints. In Sects. 4.1.1 through 4.1.4, the deformation resulting from cavity expansion is analyzed under the assumption of spherical symmetry. To capture the radially outward and circumferential nature of this deformation, the governing equations are formulated in spherical polar coordinates (r, Θ, and ϕ). This coordinate framework is particularly well suited for problems dominated by radial geometry, as it aligns with the physical symmetry of the expanding cavity. The adoption of spherical coordinates not only simplifies the derivation of equilibrium and constitutive relations but also reflects the classical analytical approach commonly used in modeling thick-walled spherical systems, as exemplified in [31].

4.1 Phenomenological models

4.1.1 Mooney–Rivlin model

Extending the Neo-Hookean framework, the Mooney–Rivlin model incorporates additional terms to accommodate large deformations [35]. By introducing two parameters to capture material behavior, this model is generally expressed as

$$W=C_{10}\left(I_1-3\right)+C_{01}(I_2-3) \tag{11}$$

Here, C_{10} and C_{01} are material constants, while, I_1 and I_2 are the invariants of the Green deformation tensor, defined in terms of the principal stretch ratios λ_1, λ_2, λ_3, $I_1=\lambda_1{}^2+\lambda_2{}^2+\lambda_3{}^2$, and $I_2=\lambda_1{}^2\lambda_2{}^2+\lambda_2{}^2\lambda_3{}^2+\lambda_3{}^2\lambda_1{}^2$. A detailed derivation for principal stretches in polar coordinates can be found in [31].

4.1.2 Yeoh model

The Yeoh model, defined as a third-order polynomial in the first invariant I_1, has proven especially effective for rubber-like materials [36]. In particular, it provides robust characterization of nonlinear load–deformation behavior over a broad strain range, while necessitating relatively minimal experimental data [37]. The SEF is generally expressed as

$$W = \sum_{i=1}^{3} C_{i0}(I_1 - 3)^i \tag{12}$$

where C_{i0} are material parameters.

4.1.3 Ogden model

The Ogden model characterizes deformation based on principal stretches rather than invariants. It captures material stiffening trends over large deformations; however, caution is warranted in the presence of limited test data, as negative material parameters may give rise to stability issues [25]. The model is expressed as

$$W = \sum_{i=1}^{n} \frac{2\boldsymbol{\mu}_i}{\boldsymbol{\alpha}_i}(\lambda_1^{\alpha_i} + \lambda_2^{\alpha_i} + \lambda_3^{\alpha_i} - 3) \tag{13}$$

where μ_i and α_i are material parameters. In the incompressible limit, the principal stretches λ_i represent the deviatoric part of the deformation.

4.1.4 Gent model

The Gent model introduces a limiting chain extensibility parameter to describe polymer behavior approaching ultimate stretch limits. This model is particularly effective in capturing stiffening effects at high deformations:

$$W = -\frac{\mu J_m}{2}\ln(1 - \frac{I_1 - 3}{J_m}) \tag{14}$$

where μ is the shear modulus, and J_m is the chain extensibility parameter.

4.2 Semi-structural models

4.2.1 Arruda–Boyce model

This model, also known as the 8-chain model, uses a molecular chain network approach with a representative hexahedron volume [38]. Although it is based on I_1 and generally performs well with minimal experimental data, its calibration efficiency in cavitation rheology is somewhat lower than that of the Yeoh and Ogden models [7]. The SEF is given by:

$$W = \mu[\frac{I_1 - 3}{2} + \frac{I_1^2 - 9}{20N} + \frac{11(I_1^3 - 27)}{1050N^2} + \frac{19(I_1^4 - 81)}{7000N^3} + \frac{519(I_1^5 - 243)}{673750N^4}] \tag{15}$$

where μ, given by $(\frac{r_0}{r_1}\lambda_m)^2$, is the initial shear modulus, and λ_m represents the extensibility limiter.

Interpreting cavitation rheology data requires advanced SEFs that extend beyond the limitations of the Neo-Hookean model. Phenomenological models such as Mooney–Rivlin, Yeoh, and Ogden excel in capturing nonlinear material responses and stiffening behavior, while semi-structural models such as Arruda–Boyce offer insights into molecular chain networks. The Gent model further accommodates finite chain extensibility effects. Collectively, these SEFs provide robust frameworks for characterizing the complex mechanical behaviors of soft materials, enabling researchers to tailor their analyses to specific applications.

5 Enhancing cavitation rheology analysis through computational simulations

Computational simulations play a pivotal role in advancing cavitation rheology by offering perspectives that are often beyond the reach of experimental methods. Techniques such as FEM and computational fluid dynamics (CFD) enable the interpretation of complex experimental data and the prediction of material behavior under diverse conditions. This section highlights the critical contributions of computational approaches to understanding cavitation phenomena.

5.1 Modeling nonlinear material behavior

Soft materials, including hydrogels, biological tissues, and polymers, frequently display nonlinear mechanical responses during cavitation. Governed by hyperelasticity and viscoelasticity, these responses pose significant challenges when experimental approaches alone are used. Numerical simulations address this gap by integrating established constitutive models to replicate such behaviors with precision. Calibrating these models against empirical data offers refined elucidations of deformation mechanics, stress distribution, and critical cavitation thresholds. For example, in VCCE experiments, simulations replicate the expansion of cavities, facilitating the quantitative extraction of mechanical properties such as elastic modulus and yield stress from cavitation pressure measurements [10, 39]. Moreover, computational analyses offer a deeper understanding of stress–strain behavior across varying loading conditions, enhancing the reliability and interpretability of experimental findings.

Beyond their role in parameter extraction, numerical methods enable rigorous exploration of cavitation behavior in regimes where analytical solutions either fail or become unreliable. Zheng and Cai [27] established clear quantitative boundaries for the validity of analytical models by comparing perturbation-based solutions for slightly compressible neo-Hookean solids against full numerical simulations. Their findings revealed that analytical predictions lose accuracy as the compressibility parameter increases ($\varepsilon > 0.2$) or when large cavity expansions occur [27], highlighting the necessity of numerical approaches in such conditions. Complementing this, Aghayan et al. [40] demonstrated, through a coupled experimental–computational approach, that numerical simulations are essential for resolving the strongly nonlinear evolution of cavitation under dynamic loading in agarose gels, particularly capturing the initiation and propagation of cavities in response to high strain-rate compressive impacts. Their study integrated high-speed imaging with FEM-based transient simulations, providing a mechanistic understanding of cavity nucleation, growth, and collapse under rapidly evolving stress fields, processes that are intractable using static models alone [40]. Mishra et al. [10] addressed real-world boundary constraints by simulating cavitation within finite-sized, viscoelastic samples [10]. Their analysis demonstrated that cavity growth in the vicinity of rigid boundaries leads to pronounced stress asymmetry and altered deformation fields, deviating from the predictions of infinite or symmetric geometries. Notably, when the cavity was positioned close to a rigid boundary, at a distance of just 300 μm, the extracted Young's modulus was overestimated by approximately 40% relative to centrally positioned cavities. Additionally, their parametric studies on sample thickness revealed that thinner specimens exhibited constrained cavity expansion and higher apparent stiffness due to geometric confinement. These findings quantitatively highlight how proximity to boundaries and specimen geometry can significantly distort both the mechanical response during cavity evolution and the accuracy of property extraction. A more detailed theoretical analysis of boundary effects and their mechanistic implications on cavity evolution is presented in the following section.

5.2 Geometrical and boundary effects

Cavitation rheology exhibits acute sensitivity to both the geometric configuration of the probed domain and the mechanical boundary conditions enforced at its periphery. These boundaries modulate the distribution of stress fields that evolve in response to cavity expansion, thereby governing the critical pressure required for cavitation onset as well as the subsequent deformation kinetics. While most theoretical frameworks adopt idealized boundary constraints to preserve mathematical tractability, such simplifications are often misaligned with the complexities inherent in experimental systems or biological tissues. In this context, we delineate three canonical boundary condition archetypes, stress-free boundaries, rigid confinement, and remote traction fields, and examine their respective influence on cavity evolution in soft elastic media.

The stress-free regime typifies an unbounded or weakly constrained medium, in which the surrounding material is permitted to deform without externally applied loads at the boundary. This idealization allows for an analytical treatment wherein the radial stress diminishes asymptotically, simplifying the derivation of pressure–volume relations. For

instance, Kang et al. (2017) employed this construct to interrogate the cavitation-to-fracture transition in elastomeric solids subjected to hydrostatic tension [41]. By modeling the surrounding matrix as an infinite, unconstrained continuum, their analysis isolated the intrinsic failure mechanisms of the material while intentionally excluding external confinement effects, an assumption appropriate for large tissue phantoms or thick hydrogel slabs commonly encountered in vitro.

By contrast, numerous experimental and physiological systems are characterized by rigid confinement, wherein the deformable medium is embedded within or constrained by a mechanically dominant boundary, such as an incompressible substrate, a stiff container, or a structural fixture. This condition is particularly salient in biomedical settings, including tumor-mimetic hydrogels cast in containment molds or soft tissues integrated within mechanically reinforced scaffolds. Hao et al. (2023) systematically examined such a scenario by modeling a compliant elastomer layer confined between two rigid plates [42]. Their analysis revealed a pronounced increase in the critical pressure required to initiate and propagate cavity formation, diverging markedly from classical models predicated on unconstrained expansion. These results underscore the imperative to account for confinement-induced stress amplification when interpreting cavitation behavior in mechanically bounded domains, particularly where boundary–material interactions fundamentally alter the energetic landscape of failure initiation.

A third frequently employed boundary condition is the far-field traction regime, in which the deformable medium is idealized as an unbounded continuum with vanishing external traction at its asymptotic boundary. This condition is particularly pertinent in the modeling of in vivo and ex vivo soft tissue systems, where the material of interest is mechanically embedded within expansive structures that do not impose localized constraints. Li et al. (2024) adopted this framework in their formulation of cylindrical cavity expansion to characterize soft tissue analogs [43]. By modeling the specimen as a thick-walled cylinder subjected to internal pressure at the cavity surface and allowing the outer boundary to extend to infinity, they analytically enforced a traction-free condition at the far field. This abstraction enabled the isolation of the local material response near the cavity, eliminating confounding effects associated with finite sample geometry or boundary-induced stress reflections.

The limiting case $\lambda_b \to 1$, derived from the geometric limit $B/A \to \infty$, encapsulates this boundary condition, serving as a critical constraint in their predictive modeling of cavity growth dynamics. Although both stress-free and far-field traction conditions imply zero stress at the boundary, they diverge in their physical and mathematical implications: the former applies to a finite, explicitly modeled surface where sample geometry remains influential, whereas the latter abstracts the boundary to infinity, effectively nullifying geometric confinement and rendering the surrounding domain mechanically inert. This distinction is essential in contexts where the goal is to capture the intrinsic material response devoid of extrinsic structural modulation. To support the discussion of boundary conditions and assumptions, Fig. 3 provides a schematic representation of a typical cavitation rheology setup.

The configuration includes an expanding cavity (e.g., fluid-filled or balloon-based) embedded within a soft material domain and subject to different boundary conditions. These include a rigid outer wall, a stress-free boundary, or an effectively infinite domain subjected to far-field traction.

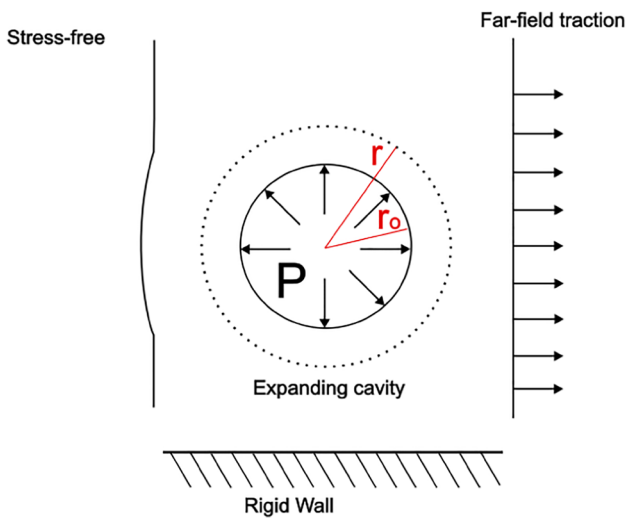


Fig. 3 Schematic illustration of a typical cavitation rheology configuration used to characterize soft material mechanics. The central pressurized cavity (radius r_o) expands radially under internal pressure P, deforming the surrounding soft solid domain. Three types of outer boundary conditions are illustrated: (i) stress-free, allowing unrestricted deformation; (ii) far-field traction, representing remote mechanical loading in an extended domain; (iii) rigid confinement, which restricts deformation at the boundary

5.3 Parametric sensitivity and experimental optimization

Beyond data interpretation, computational simulations are instrumental in optimizing experimental protocols. They allow for the exploration of a wide array of scenarios, thereby minimizing the necessity for extensive physical trials and associated costs. Parametric studies have identified critical factors influencing cavitation mechanics. Mishra et al. [10] investigated the elasto-capillary number (ECN), defined as $\frac{2w}{Er_0}$, where w is the surface tension, E is the material's elastic modulus, and r_0 is the needle size. Their findings indicated that smaller needles (higher ECN values) increase radial and hoop stresses near the needle tip, enhancing resistance to cavitation. Conversely, larger needles (lower ECN values) facilitate easier cavity expansion by distributing stresses more evenly.

Similarly, Barney and colleagues employed numerical simulations to determine the critical retraction distance (d^*_{ret}) in NIC experiments. Retracting the needle beyond d^*_{ret} was shown to minimize residual strain effects at the needle tip, enhancing the reliability of cavitation pressure measurements [44]. The simulations identified two key factors governing d^*_{ret}–(i) puncture and fracture propagation: the energy required to initiate and propagate fractures, influenced by material properties such as elastic modulus and fracture toughness, as well as needle geometry; (ii) adhesive peel forces: resistance at the needle-material interface during retraction, contributing to residual strain. By optimizing the needle retraction protocol, simulations ensured the formation of well-defined cavities, which reduced experimental variability and improved measurement accuracy.

The integration of computational simulations with experimental cavitation rheology enhances the precision and reliability of material characterization. By modeling nonlinear material behavior, accounting for geometrical and boundary effects, and optimizing experimental parameters, simulations expand the scope and accuracy of cavitation studies. These tools enable researchers to refine methodologies and deepen their understanding of the mechanical properties of soft materials, solidifying their importance in advancing the field.

5.4 Numerical modeling and dynamic characterization of LIC

LIC introduces unique opportunities and challenges in soft material characterization due to its inherently high strain rates and localized energy deposition. Unlike quasi-static methods, LIC operates in regimes dominated by inertial forces, transient pressure gradients, and rapid interface deformations. Hariri et al. [45] implemented OpenFOAM-based simulations to investigate LIC bubble dynamics in both Newtonian and non-Newtonian liquids confined within rigid cylindrical chambers. Their model resolved the formation and collapse of vapor cavities and the subsequent development of liquid microjets. Notably, simulations predicted high-velocity jets (up to ~ 28 m/s) impacting adjacent surfaces, with dynamics strongly modulated by the liquid's rheology. Moreover, their approach enabled a detailed reconstruction of the pressure distribution and jet formation pathways, elucidating how non-Newtonian viscosity modulates jet shape, collapse symmetry, and energy dissipation during bubble collapse [45].

In a rigorous theoretical and experimental study, Yang et al. [46] established a novel perturbation-based framework to predict the onset and evolution of nonspherical instabilities in inertial cavitation bubbles within nonlinear viscoelastic solids [46]. Departing from the classical spherical symmetry assumption, their model captures shape perturbations using spherical harmonic modes and explicitly incorporates nonlinear strain-stiffening behavior and viscoelastic damping. The analysis demonstrated that the circumferential stretch ratio at the bubble wall ($\lambda_w = R/R_0$) is a critical parameter governing instability onset. Instabilities were shown to emerge when $\lambda_w > 1$, particularly during rapid collapse phases where inertial effects dominate. Through coupling with high-speed imaging of laser-induced cavitation in polyacrylamide hydrogels, the model accurately reproduced experimentally observed mode transitions, including stellate patterns and localized creases, validating its predictive capability. Quantitative criteria were derived based on the signs of two stability parameters, η and ξ which depend on the bubble's acceleration, the viscoelastic material's constitutive parameters (e.g., ground-state shear modulus G, strain-stiffening coefficient α, and viscosity μ), and surface tension γ. These results not only offer a mechanistic understanding of how microscale instabilities amplify stress and strain concentrations near the bubble wall but also establish predictive thresholds for dynamic material failure, with direct implications for tissue damage in biomedical applications such as laser surgery and histotripsy.

6 Challenges and limitations in experimental setups for cavitation rheology

While cavitation rheology offers a robust framework for characterizing the mechanical properties of soft materials, experimental techniques such as NIC and VCCE face notable challenges that affect the accuracy and reliability of their results.

NIC is based on inserting a needle into the material, applying pressure to induce cavitation, and using the critical cavitation pressure to estimate material properties. Despite its utility, the method encounters several limitations, including: (i) residual strain effects–needle insertion frequently introduces residual strains into the material, leading to overestimated mechanical properties. Barney et al. [15] reported that NIC protocols often yield higher modulus values compared to conventional rheometry due to altered stress distributions around the needle. Factors such as insertion speed, depth, and the material's compliance exacerbate these effects, further distorting the measured modulus [44]; (ii) limitations of constitutive models–NIC commonly relies on the neo-Hookean constitutive model to interpret results. However, this approach is insufficient for materials with strain-stiffening behavior, such as brain tissue. Researchers demonstrated that NIC underestimates the elastic modulus of nonlinear materials, as the neo-Hookean model assumes a linear response. This leads to inconsistencies in characterizing biological tissues, such as liver or muscle, where stiffness increases significantly under strain [24]; (iii) cavitation instability–NIC assumes cavitation occurs once the critical pressure threshold is reached. However, in strain-stiffening materials, cavitation may be suppressed or delayed. Analytical models indicate that such materials sustain higher pressures before cavitation initiates, as increased stiffness counteracts bubble formation. Consequently, NIC overestimates cavitation thresholds, impacting measurement accuracy [47].

The VCCE method, which controls cavity expansion volume, also faces unique challenges, including (i) strain rate sensitivity–many soft materials exhibit viscoelastic behavior, where mechanical responses vary with the deformation rate. Nafo and Al-Mayah observed that strain rate variations significantly influenced the response of PVA hydrogels, indicating the need to account for viscoelastic effects in VCCE measurements [7]. Similarly, Chockalingam et al. [9] noted that neglecting strain rate sensitivity can lead to unreliable results, as materials may stiffen or soften depending on the loading rate [9]; (ii) cavity shape assumptions–VCCE often assumes that the expanding cavity remains spherical. While valid for isotropic materials, this assumption breaks down in anisotropic or heterogeneous materials. Nafo and Al-Mayah emphasized the importance of validating cavity shape with imaging techniques such as computed tomography (CT), as deviations from sphericity alter stress distributions and measurement accuracy [11]; (iii) boundary proximity significantly impacts stress distributions around expanding cavities, particularly near rigid interfaces. Such constraints can result in non-uniform stress fields and an apparent increase in material stiffness. Liu and colleagues demonstrated that positioning cavities far from boundaries is essential to minimize boundary-induced errors and ensure accurate material characterization [48].

Both NIC and VCCE provide valuable tools for probing the mechanical properties of soft materials; however, their accuracy depends on addressing inherent methodological limitations. Residual strain effects, geometry-based assumptions, and boundary influences must be carefully accounted for to ensure reliable measurements. Advancing cavitation rheology requires the integration of improved experimental protocols, high-resolution imaging techniques, and computational modeling to systematically mitigate these challenges. By adopting such a multifaceted approach, researchers can enhance the precision and reproducibility of cavitation-based measurements, solidifying its role as a robust method for soft material characterization.

7 Tissue-specific responses to cavitation and influencing factors

Cavity expansion techniques, including NIC, VCCE, and BRC, serve as critical tools for characterizing the mechanical properties of biological tissues. These methods are based on creating a cavity within the tissue through fluid injection or mechanical expansion and analyzing the resultant pressure–volume relationship to infer material characteristics.

While cavitation rheology has been successfully applied to various soft tissues, including liver, brain, and blood clots, its utility depends strongly on the mechanical architecture and functional properties of the tissue in question. Brain and liver tissues are particularly well-suited to this technique, as they are largely homogeneous (at the scale probed), and viscoelastic in behavior. These mechanical characteristics align with the assumptions underpinning cavitation analysis. For instance, Mijailovic et al. [24] demonstrated that needle-induced cavitation rheology yields consistent elastic moduli across different brain regions, closely matching results from complementary mechanical tests [24].

By contrast, highly anisotropic or active tissues, notably the heart, present significant challenges. Cardiac tissue is composed of aligned muscle fibers and exhibits active contraction, fiber reinforcement, and inherent pre-stress. These features violate key assumptions of cavitation analysis, such as isotropy, passive response, and uniform boundary conditions. Indeed, anisotropic electromechanical models of cardiac tissue (e.g., [49]) include orthotropic elasticity and active stress components essential for accurate simulation [49]. Such complexity severely limits the applicability of cavitation techniques unless paired with advanced constitutive models or imaging to resolve local fiber orientations and boundary forces.

Tendons, ligaments, and highly fibrous tissues also resist traditional cavitation rheology applications, as the presence of collagen networks introduces non-uniform, strain-stiffened behavior and prevents the cavity from expanding in a symmetric

fashion. These tissues require alignment-sensitive mechanical protocols or combinations with shear-based or tensile testing to capture their hierarchical structure.

Liver tissue, characterized by its soft and homogeneous structure, exhibits a relatively uniform mechanical response to cavity expansion, making it well-suited for assessing nonlinear elastic properties under VCCE protocols. These protocols enable precise quantification of evolving tissue stiffness by analyzing pressure–volume relationships over time, supported by real-time imaging and high-precision measurement techniques. Such methodologies are critical for investigating viscoelastic alterations associated with pathological conditions, including fibrosis and necrosis. However, replicating in vivo conditions while minimizing tissue damage during experimentation remains a significant challenge. Advances in computational modeling and refinements in experimental design are continuously addressing these limitations, improving both physiological relevance and measurement accuracy [50]. Porcine liver, commonly utilized as a structural analog for human liver, has been extensively studied using the BRC method. Nafo and Al-Mayah investigated the hyperelastic response of porcine liver by embedding and inflating a balloon within the tissue, enabling the characterization of stress–strain behavior and the determination of key mechanical properties, including the elastic modulus and strain-hardening response. Their findings highlighted the significant influence of experimental parameters—such as balloon size, inflation rate, and tissue heterogeneity—on measurement accuracy [25].

Brain tissues, with their delicate and heterogeneous microarchitecture, pose unique challenges for cavity expansion methods. Needle-induced cavitation requires meticulous control to prevent structural damage, which could compromise measurement accuracy. The technique involves the injection of controlled fluid volumes to create cavities while minimizing stress concentrations that might disrupt the fragile tissue structure. Recent advancements, including the use of finer needle gauges, real-time imaging, and computational frameworks, have significantly enhanced experimental precision. These innovations allow researchers to derive localized mechanical properties of brain tissue, crucial for understanding traumatic brain injuries and developing therapeutic strategies [51].

Blood clots exhibit a complex mechanical response to cavity expansion due to their viscoelastic nature, which governs the balance between elastic deformation and viscous dissipation. VCCE techniques have been employed to characterize clot mechanics by analyzing pressure variations during expansion. Key parameters, including the storage modulus and loss modulus, reflect clot stability and dynamic behavior under physiological conditions. These measurements are critical for modeling clot mechanics, advancing the understanding of thrombotic diseases, and optimizing anticoagulant therapies. However, replicating in vivo conditions, such as blood flow and temperature, remains a significant challenge, as these factors strongly influence clot properties. To address these limitations, researchers are integrating bioreactors and advanced imaging systems into experimental protocols to enhance physiological relevance and measurement accuracy [8].

The response of biological tissues to cavity expansion is highly tissue-specific, shaped by factors such as structural composition, mechanical properties, and susceptibility to damage. Techniques such as NIC, VCCE, and BRC must be tailored to the unique characteristics of each tissue type to ensure accurate characterization. Advances in imaging, computational modeling, and experimental precision continue to refine these methods, enhancing their applicability in medical diagnostics and therapeutic developments. These innovations not only improve our understanding of tissue mechanics but also hold significant implications for advancing clinical care and treatment strategies.

8 Integrating cavitation rheology with complementary biomechanical testing methods

When integrating cavitation rheology with other biomechanical testing methods, it provides a more comprehensive understanding of tissue mechanics across scales, bridging localized measurements with bulk material behavior. Below, we explore key integrations that enhance the study of tissue and soft material mechanics.

8.1 Combining cavitation rheology with tensile and fracture testing

The integration of cavitation rheology with tensile and fracture testing offers a comprehensive framework for investigating the mechanical behavior of physically assembled polymer gels across varying polymer concentrations [52]. Cavitation rheology excels in probing localized mechanical properties, particularly in low-modulus gels, by quantifying critical energy release rates (Γ_0) through controlled-pressure cavity expansion. For higher-modulus gels, tensile testing captures rate-dependent stress–strain behavior and strain-enhanced modulus, attributed to polymer entanglement effects. This integration emphasizes the interplay between localized fracture phenomena and global mechanical integrity, bridging

the gap between localized assessments and bulk mechanical characterizations. Together, these methods enable a robust analysis of soft material mechanics across different deformation modes and polymer concentrations.

8.2 Integration with shear rheology and dynamic low-velocity impact

Ji and colleagues explored the mechanical behavior of triblock copolymer gels by integrating shear rheology, cavitation rheology, and dynamic low-velocity impact testing, focusing on strain-rate dependencies, gel stiffness across varying polymer concentrations, and practical stiffness measurement strategies [53].

Shear rheology, conducted under controlled strain rates, revealed the viscoelastic properties of the gels, with a consistent increase in stiffness observed as polymer concentration increased. Cavitation rheology complemented these findings by providing localized stiffness measurements via cavity expansion, effectively capturing Young's modulus at low strain rates. This was particularly evident for softer gels with lower polymer concentrations.

Dynamic low-velocity impact testing further explained strain-rate effects, demonstrating a rate-dependent modulus increase in stiffer gels. Softer gels, in contrast, exhibited deformation phenomena such as pinch-off, aligning with cavitation rheology results.

Together, these techniques present a robust and synergistic framework for analyzing gel stiffness, integrating localized and bulk measurements to capture strain-rate effects and material responses across a broad range of polymer concentrations. This comprehensive approach emphasizes the utility of pragmatic methodologies for characterizing soft materials under diverse mechanical conditions.

8.3 Integration with shear wave elastography

Researchers investigated the interaction between cavitation dynamics and shear wave generation in soft solids, using gelatin as a representative tissue-mimicking material [54]. Their study demonstrated that cavitation bubbles, induced by laser pulses, generate shear waves during asymmetric bubble collapse near a free surface. This process is governed by the material's elastic restoring forces, which confine shear stresses at the interface between the bubble and the air–solid boundary. These stresses are subsequently released as propagating shear waves, with velocities dependent on the material's shear modulus and density. Gelatin, characterized by its viscoelastic nature, exhibits both elastic and viscous responses under deformation, allowing it to support the transmission of both longitudinal and shear waves. During bubble collapse, localized material deformation leads to stress patterns that are captured using photoelastic imaging, revealing birefringent structures indicative of shear wave generation. This highlights the material's ability to concentrate and transmit mechanical energy, a critical feature for understanding tissue mechanics.

Shear wave propagation is further influenced by the anisotropy and heterogeneity of the material. In gelatin, shear wave velocities (1–50 m/s) are significantly lower than those of longitudinal waves (~ 1500 m/s), emphasizing the material's soft and compliant nature. The asymmetric collapse of cavitation bubbles also contributes to directional shear wave emission, mimicking real-world tissue behavior during cavitation events, as observed in medical applications such as histotripsy.

These findings highlight the potential of integrating cavitation rheology with shear wave elastography to achieve a dual-perspective assessment of material behavior. Cavitation rheology enables site-specific characterization of stress–strain responses by probing localized mechanical properties at the point of bubble collapse. In contrast, shear wave elastography provides macroscopic stiffness measurements through propagating wave-based analysis. Together, these complementary techniques enhance the understanding of soft tissue mechanics, offering improved diagnostic precision and informing therapeutic interventions in applications such as ultrasound imaging, laser surgery, and tissue ablation.

Integrating cavitation rheology with complementary biomechanical testing methods provides a comprehensive framework for understanding soft material mechanics. By combining localized stress–strain analysis with bulk property assessments, such as tensile and fracture testing, shear rheology, and dynamic impact testing, researchers can bridge the gap between localized and global material behavior. The addition of shear wave elastography and ultrasound imaging further extends this framework, capturing wave-based stiffness metrics and dynamic responses. These integrations refine the understanding of anisotropy, heterogeneity, and tissue mechanics, enhancing diagnostic and therapeutic applications. Together, these approaches establish cavitation rheology as a vital tool in soft material and tissue characterization.

9 Limitations and future directions

Despite its utility, cavitation rheology continues to face Fundamental limitations that must be systematically addressed to broaden its applicability and precision. A primary concern is the impact of boundary conditions and geometric constraints on test outcomes. Standard cavitation models often assume an infinite homogeneous medium, but actual experiments frequently involve samples near rigid supports or small laboratory specimens. High-speed visualizations in soft elastomers have demonstrated that proximity to rigid boundaries can distort bubble collapse dynamics and elevate cavitation thresholds by up to 30% due to reflected stress waves and confinement effects not captured in ideal models.

Another critical challenge lies in the rate-dependent and irreversible nature of soft material deformation. Many soft tissues and hydrogels display viscoelastic or poroelastic behavior, and cavitation itself can induce damage zones with irreversible molecular changes. A recent study in PDMS and alginate hydrogels showed that cavity expansion triggers damage accumulation around the cavity rim, even under moderate loading rates, leading to elastic-to-fracture transitions not adequately described by traditional hyperelastic frameworks [43]. These results underscore the need for experimental setups capable of quantifying time-resolved mechanical changes, as well as constitutive models that incorporate relaxation and damage evolution.

A third limitation concerns material anisotropy, which is largely overlooked in current cavitation applications. Many tissues are fiber-reinforced (e.g., myocardium, ligaments, tendons) and exhibit strongly direction-dependent behavior. Experimental investigations in anisotropic gels and engineered tissues have revealed that bubble growth can proceed non-spherically, breaking symmetry and significantly altering local stress fields [46, 55]. Without accounting for such anisotropic effects, pressure–volume data can misrepresent intrinsic tissue properties. Future methodologies should integrate cavitation rheology with structural imaging (e.g., diffusion tensor MRI) or include anisotropic constitutive laws.

Moreover, spatial heterogeneity, common in tumors, fibrotic organs, and engineered scaffolds, presents a major blind spot. Conventional cavitation rheology yields a single local mechanical measurement, which may not reflect nearby stiffness variations. Emerging strategies using multiple cavity arrays or imaging-integrated probes promise to map mechanical heterogeneity, but they remain underdeveloped and unstandardized.

To move forward, the field should prioritize the following research directions: (1) development of fluid–structure interaction numerical models incorporating realistic boundary geometries and transient loading; (2) design of dynamic cavitation protocols, such as controlled rate expansion or laser-induced cavities, to extract viscoelastic and damage parameters; (3) fusion of cavitation testing with imaging modalities to capture anisotropy and heterogeneity in situ; (4) establishment of community benchmarks, including unified reporting standards, sample geometries, and inter-lab protocols. Addressing these interlinked challenges will be essential to elevate cavitation rheology into a widely applicable, high-resolution tool for probing the complexity of soft biological and synthetic materials.

Acknowledgements This study was financially supported by the European Commission under the Marie Skłodowska-Curie Actions, grant no. 101067240.

Author contribution Daniel Coen: Conceptualization; investigation; methodology; writing—original draft. Kaylin Shanahan: Conceptualization; investigation; methodology; writing—review and editing. Wanis Nafo: Conceptualization; funding acquisition; investigation; methodology; project administration; supervision; writing—review and editing.

Funding This study was financially supported by the European Commission under the Marie Skłodowska-Curie Actions, grant no. 101067240.
HORIZON EUROPE Framework Programme,101067240,Wanis Nafo

Availability of data and material Not applicable.

Declarations

Ethics approval and consent to participate Not applicable.

Consent for publication All authors have read and approved the final version of the manuscript and consent to its submission for publication.

Competing interests The authors declare no competing interests.

References

1. Feng, Z., Kosawada, T., Nakamura, T., et al.: Theoretical methods and models for mechanical properties of soft biomaterials. AIMS Materials Science **4**, 680–705 (2017). https://doi.org/10.3934/matersci.2017.3.680
2. Tac, V., Sree, V.D., Rausch, M.K., Tepole, A.B.: Data-driven modeling of the mechanical behavior of anisotropic soft biological tissue. Engineering with Computers **38**, 4167–4182 (2022). https://doi.org/10.1007/s00366-022-01733-3
3. Liu, S., Yu, J.-M., Gan, Y.-C., et al.: Biomimetic natural biomaterials for tissue engineering and regenerative medicine: new biosynthesis methods, recent advances, and emerging applications. Military Medical Research **10**, 16 (2023). https://doi.org/10.1186/s40779-023-00448-w
4. Hutchens, S.B., Fakhouri, S., Crosby, A.J.: Elastic cavitation and fracture via injection. Soft Matter **12**, 2557–2566 (2016). https://doi.org/10.1039/C5SM02055G
5. Fu, Y., Yin, T., Qu, S., Yang, W.: Cavitation/fracture transition of soft materials. J. Mech. Phys. Solids **172**, 105192 (2023). https://doi.org/10.1016/j.jmps.2022.105192
6. Zhang, W., Capilnasiu, A., Nordsletten, D.: Comparative analysis of nonlinear viscoelastic models across common biomechanical experiments. J. Elast. **145**, 117–152 (2021). https://doi.org/10.1007/s10659-021-09827-7
7. Nafo, W., Al-Mayah, A.: Mechanical characterization of PVA hydrogels' rate-dependent response using multi-axial loading. PLoS ONE **15**, e0233021 (2020). https://doi.org/10.1371/journal.pone.0233021
8. Varner, H., Sugerman, G.P., Rausch, M.K., Cohen, T.: Elasticity of whole blood clots measured via volume controlled cavity expansion. J. Mech. Behav. Biomed. Mater. **143**, 105901 (2023). https://doi.org/10.1016/j.jmbbm.2023.105901
9. Chockalingam, S., Roth, C., Henzel, T., Cohen, T.: Probing local nonlinear viscoelastic properties in soft materials. J. Mech. Phys. Solids **146**, 104172 (2021). https://doi.org/10.1016/j.jmps.2020.104172
10. Mishra, S., Lacy, T.E., Kundu, S.: Effect of surface tension and geometry on cavitation in soft solids. Int. J. Non-Linear Mech. **98**, 23–31 (2018). https://doi.org/10.1016/j.ijnonlinmec.2017.10.001
11. Nafo, W., Al-Mayah, A.: Measuring hyperelastic properties of hydrogels using cavity expansion method. Exp. Mech. (2019). https://doi.org/10.1007/s11340-019-00504-4
12. Zimberlin, J.A., McManus, J.J., Crosby, A.J.: Cavitation rheology of the vitreous: mechanical properties of biological tissue. Soft Matter **6**, 3632 (2010). https://doi.org/10.1039/b925407b
13. Shen, Z., Shao, J., Zhang, J., Qu, W.: Ultrasound cavitation enhanced chemotherapy: in vivo research and clinical application. Exp. Biol. Med. (Maywood) **245**, 1200–1212 (2020). https://doi.org/10.1177/1535370220936150
14. Marsh, J.L., Bentil, S.A.: Cerebrospinal fluid cavitation as a mechanism of blast-induced traumatic brain injury: a review of current debates, methods, and findings. Front. Neurol. **12**, 626393 (2021). https://doi.org/10.3389/fneur.2021.626393
15. Barney, C.W., Dougan, C.E., McLeod, K.R., et al.: Cavitation in soft matter. Proc. Natl. Acad. Sci. U. S. A. **117**, 9157–9165 (2020). https://doi.org/10.1073/pnas.1920168117
16. Zimberlin, J.A., Sanabria-DeLong, N., Tew, G.N., Crosby, A.J.: Cavitation rheology for soft materials. Soft Matter **3**, 763 (2007). https://doi.org/10.1039/b617050a
17. Delbos, A., Cui, J., Fakhouri, S., Crosby, A.J.: Cavity growth in a triblock copolymer polymer gel. Soft Matter **8**, 8204 (2012). https://doi.org/10.1039/c2sm25458a
18. Kundu, S., Crosby, A.J.: Cavitation and fracture behavior of polyacrylamide hydrogels. Soft Matter **5**, 3963 (2009). https://doi.org/10.1039/b909237d
19. Pavlovsky, L., Ganesan, M., Younger, J.G., Solomon, M.J.: Elasticity of microscale volumes of viscoelastic soft matter by cavitation rheometry. Appl. Phys. Lett. **105**, 114105 (2014). https://doi.org/10.1063/1.4896108
20. Nafo, W., Al-Mayah, A.: Characterization of PVA hydrogels' hyperelastic properties by uniaxial tension and cavity expansion tests. Int. J. Non-Linear Mech. **124**, 103515 (2020). https://doi.org/10.1016/j.ijnonlinmec.2020.103515
21. Raayai-Ardakani, S., Chen, Z., Earl, D.R., Cohen, T.: Volume-controlled cavity expansion for probing of local elastic properties in soft materials. Soft Matter **15**, 381–392 (2019). https://doi.org/10.1039/C8SM02142B
22. Luo, J.C., Ching, H., Wilson, B.G., et al.: Laser cavitation rheology for measurement of elastic moduli and failure strain within hydrogels. Sci. Rep. **10**, 13144 (2020). https://doi.org/10.1038/s41598-020-68621-y
23. Senegačnik, M., Kunimoto, K., Yamaguchi, S., et al.: Dynamics of laser-induced cavitation bubble during expansion over sharp-edge geometry submerged in liquid – an inside view by diffuse illumination. Ultrason. Sonochem. **73**, 105460 (2021). https://doi.org/10.1016/j.ultsonch.2021.105460
24. Mijailovic, A.S., Galarza, S., Raayai-Ardakani, S., et al.: Localized characterization of brain tissue mechanical properties by needle induced cavitation rheology and volume controlled cavity expansion. J. Mech. Behav. Biomed. Mater. **114**, 104168 (2021). https://doi.org/10.1016/j.jmbbm.2020.104168
25. Nafo, W., Al-Mayah, A.: Measuring the hyperelastic response of porcine liver tissues *in-vitro* using controlled cavitation rheology. Exp. Mech. **61**, 445–458 (2021). https://doi.org/10.1007/s11340-020-00674-6
26. Kim, C., Choi, W.J., Ng, Y., Kang, W.: Mechanically induced cavitation in biological systems. Life **11**, 546 (2021). https://doi.org/10.3390/life11060546
27. Zheng, Y., Cai, S.: Analytical solutions of cavitation instability in a compressible hyperelastic solid. Int. J. Non-Linear Mech. **126**, 103562 (2020). https://doi.org/10.1016/j.ijnonlinmec.2020.103562
28. Hashemnejad, S.M., Kundu, S.: Nonlinear elasticity and cavitation of a triblock copolymer gel. Soft Matter **11**, 4315–4325 (2015). https://doi.org/10.1039/C5SM00330J
29. Fei, P., Wood, S.J., Chen, Y., Cavicchi, K.A.: Maximum bubble pressure rheology of low molecular mass organogels. Langmuir **31**, 492–498 (2015). https://doi.org/10.1021/la503832r
30. Bentz, K.C., Sultan, N., Savin, D.A.: Quantitative relationship between cavitation and shear rheology. Soft Matter **14**, 8395–8400 (2018). https://doi.org/10.1039/C8SM01560K

31. deBotton, G., Bustamante, R., Dorfmann, A.: Axisymmetric bifurcations of thick spherical shells under inflation and compression. International Journal of Solids and Structures **50**, 403–413 (2013). https://doi.org/10.1016/j.ijsolstr.2012.10.004
32. Mangan, R., Destrade, M.: Gent models for the inflation of spherical balloons. Int. J. Non-Linear Mech. **68**, 52–58 (2015). https://doi.org/10.1016/j.ijnonlinmec.2014.05.016
33. Raayai-Ardakani, S., Cohen, T.: Capturing strain stiffening using volume controlled cavity expansion. Extrem. Mech. Lett. **31**, 100536 (2019). https://doi.org/10.1016/j.eml.2019.100536
34. Estrada, J.B., Barajas, C., Henann, D.L., et al.: High strain-rate soft material characterization via inertial cavitation. J. Mech. Phys. Solids **112**, 291–317 (2018). https://doi.org/10.1016/j.jmps.2017.12.006
35. Mooney, M.: A theory of large elastic deformation. J. Appl. Phys. **11**, 582–592 (1940). https://doi.org/10.1063/1.1712836
36. Yeoh, O.H.: Some forms of the strain energy function for rubber. Rubber Chem. Technol. **66**, 754–771 (1993). https://doi.org/10.5254/1.3538343
37. Shahzad, M., Kamran, A., Siddiqui, M.Z., Farhan, M.: Mechanical characterization and FE modelling of a hyperelastic material. Materials Research **18**, 918–924 (2015). https://doi.org/10.1590/1516-1439.320414
38. Arruda, E.M., Boyce, M.C.: A three-dimensional constitutive model for the large stretch behavior of rubber elastic materials. J. Mech. Phys. Solids **41**, 389–412 (1993). https://doi.org/10.1016/0022-5096(93)90013-6
39. Mohammadigoushki, H., Shoele, K.: Cavitation rheology of model yield stress fluids based on carbopol. Langmuir **39**, 7672–7683 (2023). https://doi.org/10.1021/acs.langmuir.3c00483
40. Aghayan, S., Weinberg, K.: Experimental and numerical investigation of dynamic cavitation in agarose gel as a soft tissue simulant. Mech. Mater. **175**, 104486 (2022). https://doi.org/10.1016/j.mechmat.2022.104486
41. Kang, J., Wang, C., Cai, S.: Cavitation to fracture transition in a soft solid. Soft Matter **13**, 6372–6376 (2017). https://doi.org/10.1039/C7SM01479A
42. Hao, S., Suo, Z., Huang, R.: Why does an elastomer layer confined between two rigid blocks grow numerous cavities? J. Mech. Phys. Solids **173**, 105223 (2023). https://doi.org/10.1016/j.jmps.2023.105223
43. Li, J., Xie, Z., Varner, H., et al.: Cylindrical cavity expansion: a novel method for characterizing the mechanical properties of soft materials (2024). https://doi.org/10.48550/arXiv.2408.12920
44. Barney, C.W., Zheng, Y., Wu, S., et al.: Residual strain effects in needle-induced cavitation. Soft Matter **15**, 7390–7397 (2019). https://doi.org/10.1039/C9SM01173K
45. Hariri, A., Shervani-Tabar, M.T., Parvizi, R.: Laser-produced cavitation bubble behavior in Newtonian and non-Newtonian liquid inside a rigid cylinder: numerical study of liquid disc microjet impact using OpenFOAM. Micromachines **14**, 1416 (2023). https://doi.org/10.3390/mi14071416
46. Yang, J., Tzoumaka, A., Murakami, K., et al.: Predicting complex nonspherical instability shapes of inertial cavitation bubbles in viscoelastic soft matter. Phys. Rev. E **104**, 045108 (2021). https://doi.org/10.1103/PhysRevE.104.045108
47. Raayai-Ardakani, S., Earl, D.R., Cohen, T.: The intimate relationship between cavitation and fracture. Soft Matter **15**, 4999–5005 (2019). https://doi.org/10.1039/C9SM00570F
48. Liu, J., Zhang, X., Liu, C., et al.: Study on the boundary stress distribution of elliptical cavity under uniform pressure load. IET Conference Proceedings **2022**, 986–995 (2022)
49. Propp, A., Gizzi, A., Levrero-Florencio, F., Ruiz-Baier, R.: An orthotropic electro-viscoelastic model for the heart with stress-assisted diffusion. (2019). https://doi.org/10.48550/ARXIV.1905.00262
50. Varner, H.M., Naghibzadeh, S.K., Spaeth, K.C. et al. Evolving properties of biological materials captured via needle-based cavity expansion method. Exp Mech 65, 141–155 (2025). https://doi.org/10.1007/s11340-024-01128-z
51. Dougan, C.E., Fu, H., Crosby, A.J., Peyton, S.R.: Needle-induced cavitation: a method to probe the local mechanics of brain tissue. J. Mech. Behav. Biomed. Mater. **160**, 106698 (2024). https://doi.org/10.1016/j.jmbbm.2024.106698
52. Mishra S, Prado RMB, Kundu S (2020) Probing large deformation and fracture behavior of physically assembled gel system by varying polymer concentration
53. Ji, Y., Dagro, A.M., Dorgant, G., et al.: A comparison of conventional gel stiffness characterization techniques with cavitation rheology. Exp. Mech. **62**, 799–812 (2022). https://doi.org/10.1007/s11340-022-00829-7
54. Rapet J, Tagawa Y, Ohl C-D (2019) Shear-wave generation from cavitation in soft solids. https://doi.org/10.48550/ARXIV.1902.02167
55. Long, R., Hui, C.-Y., Gong, J.P., Bouchbinder, E.: The fracture of highly deformable soft materials: a tale of two length scales. (2020). https://doi.org/10.48550/ARXIV.2004.03159

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.